Complex numbers on the unit circle

Kevin Hu
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L5-X5 Level 5 Advanced · Geometry · Beyond the textbook

The unit circle set-up, formulas for chords, tangents, feet and reflections, the conjugate trick, the square-root substitution for the incentre, and how to write a complex bash that earns full marks.
By the end you can place a configuration on the unit circle and write every point with the standard formulas; prove collinearity, perpendicularity and concyclicity by the conjugate trick, including the incentre via a = x squared; judge when a complex bash will finish, and write one a marker will accept.
Prerequisites: Olympiad angle chasing (L4-GE1), Lengths and ratios (L4-GE2). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

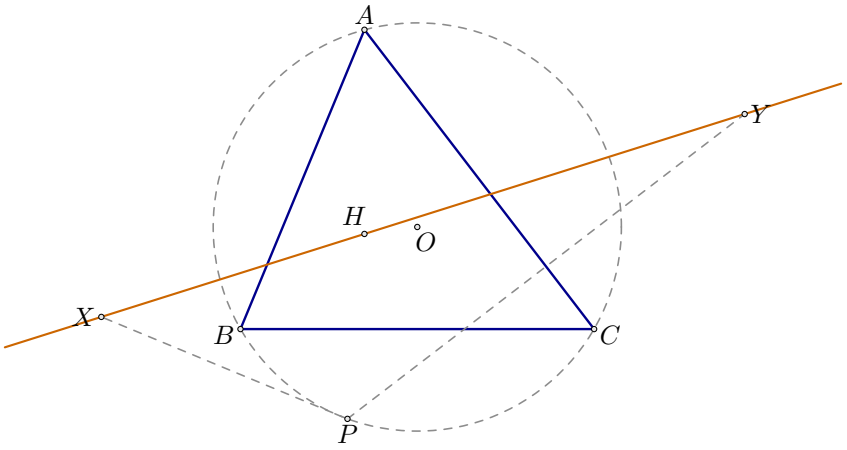
Let P be a point on the circumcircle of triangle ABC, and let X and Y be the reflections of P in the lines AB and AC. Prove that line XY passes through the orthocentre H. Take the circumcircle as the unit circle and write each point as a complex number with the same lower-case letter. Theorem 2.1 below gives $x = a + b - ab/p$, $y = a + c - ac/p$ and $h = a + b + c$. Hence $x - h = -(ab + cp)/p$ and $y - h = -(ac + bp)/p$, so

$$\frac{x-h}{y-h} = \frac{ab+cp}{ac+bp}.$$

Replacing each of a, b, c, p by its reciprocal, which is its conjugate, gives $\frac{\frac{1}{ab}+\frac{1}{cp}}{\frac{1}{ac}+\frac{1}{bp}} = \frac{(cp+ab)/abcp}{(bp+ac)/abcp}$, the same number. A complex number equal to its conjugate is real, so X, Y, H are collinear. (If $ac + bp = 0$ then $y = h$ and there is nothing to prove.) This is the Steiner line of P.

That is the whole method. Put the important circle on the unit circle, write every point as a rational expression in a few variables of modulus 1, and verify the conclusion by taking conjugates. The method is olympiad folklore. The standard references are T. Andreescu and D. Andrica, *Complex Numbers from A to ... Z*, and Chapter 6 of Evan Chen's *Euclidean Geometry in Mathematical Olympiads*. School courses treat complex numbers through modulus and argument and never use the identity $\bar{z} = 1/z$ for $|z| = 1$, which is what makes the unit circle special. Angle chasing and lengths (L4-GE1, L4-GE2) are assumed.

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1.1 The dictionary

Theorem 1.1.
(1) z is real if and only if $z = \bar{z}$, and purely imaginary if and only if $z = -\bar{z}$. (2) $|z| = 1$ if and only if $\bar{z} = 1/z$. (3) For distinct points, A, B, C are collinear if and only if $\frac{a-b}{a-c}$ is real, $AB \parallel CD$ if and only if $\frac{a-b}{c-d}$ is real, and $AB \perp CD$ if and only if $\frac{a-b}{c-d}$ is purely imaginary. (4) Four distinct points A, B, C, D lie on a circle or a line if and only if $\frac{(a-c)(b-d)}{(a-d)(b-c)}$ is real.

Statements (1) and (2) are $z\bar{z} = |z|^2$. For (3), the argument of $\frac{a-b}{c-d}$ is the angle from DC to BA. For (4), the argument of the quotient is $\angle ACB - \angle ADB$ as directed angles, and this is 0 or 180° exactly in the concyclic or collinear case. From now on the circle in question is the unit circle with centre 0.

1.2 Formulas on the unit circle

Theorem 2.1.
Let a, b, c, d lie on the unit circle and let p be any point.
(1) The line AB is $\{z : z + ab\bar{z} = a + b\}$.
(2) The tangent at A is $z + a^2\bar{z} = 2a$, and the tangents at A and B meet at $\frac{2ab}{a+b}$.
(3) If $AB \nparallel CD$, the lines AB and CD meet at $\frac{ab(c+d)-cd(a+b)}{ab-cd}$.
(4) The foot of the perpendicular from P to AB is $\frac{1}{2}(a + b + p - ab\bar{p})$.
(5) The reflection of P in AB is $a + b - ab\bar{p}$.
(6) $AB \parallel CD$ if and only if $ab = cd$, and $AB \perp CD$ if and only if $ab + cd = 0$.
(7) The centroid is $\frac{1}{3}(a + b + c)$, the orthocentre is $h = a + b + c$, and the nine-point centre is $\frac{1}{2}(a + b + c)$.

Proof. (1) z is on AB if and only if $\frac{z-a}{b-a}$ equals its conjugate $\frac{\bar{z}-1/a}{1/b-1/a} = \frac{-b(a\bar{z}-1)}{b-a}$, that is, $z - a = -ab\bar{z} + b$. (2) The tangent at A is perpendicular to OA, so $\frac{z-a}{a}$ is purely imaginary, that is, $z - a = -a^2(\bar{z} - \frac{1}{a})$. Subtracting the equations for A and B gives $(a^2 - b^2)\bar{z} = 2(a - b)$, so $\bar{z} = \frac{2}{a+b}$ and $z = 2a - \frac{2a^2}{a+b} = \frac{2ab}{a+b}$. (3) Subtracting the equations of the two lines gives $(ab - cd)\bar{z} = a + b - c - d$. Then $z = a + b - ab\bar{z}$ simplifies to the formula. (4) The conjugate of $a - b$ is $-\frac{a-b}{ab}$, so the line through P perpendicular to AB is $z - p = ab(\bar{z} - \bar{p})$. Adding this to (1) gives $2z = a + b + p - ab\bar{p}$. (5) is twice the foot minus p. (6) The conjugate of $\frac{a-b}{c-d}$ is $\frac{(a-b)cd}{(c-d)ab}$, which equals $\pm\frac{a-b}{c-d}$ exactly when

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$cd = \pm ab$. (7) With $h = a + b + c$, $\frac{h-a}{b-c} = \frac{b+c}{b-c}$ has conjugate $\frac{1/b+1/c}{1/b-1/c} = -\frac{b+c}{b-c}$, so $AH \perp BC$, and similarly for the other altitudes. The nine-point centre is the midpoint of OH. ■

Formulas (1) to (6) need a, b, c, d on the unit circle, while p is arbitrary, and $\bar{p} = 1/p$ only if P is on the circle too.

1.3 The conjugate trick

To show that an expression w is real, compute $\bar{w}$ by replacing every variable on the unit circle by its reciprocal, clear the denominators by multiplying by the product of the variables, and compare with w. Two facts shorten most computations. First, for a, b on the unit circle the number $\frac{a-b}{a+b}$ is purely imaginary, since its conjugate is $\frac{b-a}{b+a}$. A product of two purely imaginary numbers is real. Second, lengths are handled through squares, since $|a-b|^2 = (a-b)(\frac{1}{a} - \frac{1}{b}) = -\frac{(a-b)^2}{ab}$.

Example 1.
Let P be a point on the circumcircle of triangle ABC, other than A, B, C, and let X_A, X_B, X_C be the feet of the perpendiculars from P to BC, CA, AB. Prove that X_A, X_B, X_C lie on a line (the Simson line of P), that this line passes through the midpoint M of PH, and that the Simson lines of two different points P, Q of the circle meet at $\frac{1}{2}\left(p + q + a + b + c + \frac{abc}{pq}\right)$.

Solution.
By Theorem 2.1(4) with $\bar{p} = 1/p$,

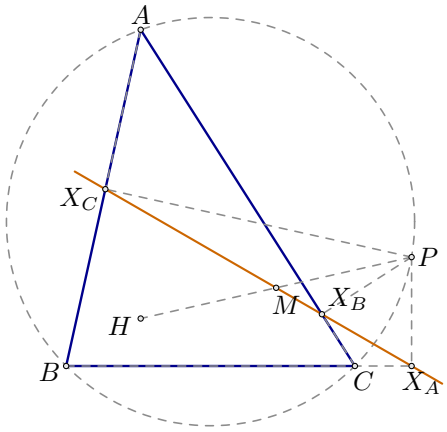
$$x_A = \tfrac{1}{2}\left(p + b + c - \tfrac{bc}{p}\right), \quad x_B = \tfrac{1}{2}\left(p + c + a - \tfrac{ca}{p}\right), \quad x_C = \tfrac{1}{2}\left(p + a + b - \tfrac{ab}{p}\right).$$

Hence $x_A - x_C = \frac{(c-a)(p-b)}{2p}$ and $x_B - x_C = \frac{(c-b)(p-a)}{2p}$, and

$$\frac{x_A - x_C}{x_B - x_C} = \frac{(c-a)(p-b)}{(c-b)(p-a)}.$$

This is the cross-ratio of four points of the circle, real by Theorem 1.1(4). So the feet are collinear. (If $x_B = x_C$, which needs $p = a$, the statement is empty. We excluded that case.)
The direction $d = x_A - x_C$ satisfies $\bar{d} = \frac{(a-c)(b-p)}{2abc}$, so $d/\bar{d} = abc/p$. A point z lies on the Simson line if and only if $z - x_C = \frac{abc}{p}(\bar{z} - \bar{x}_C)$. For $m = \frac{1}{2}(p+h)$ we get $m - x_C = \frac{1}{2}\left(c + \frac{ab}{p}\right)$, and $\frac{abc}{p} \cdot \frac{1}{2}\left(\frac{1}{c} + \frac{p}{ab}\right) = \frac{1}{2}\left(\frac{ab}{p} + c\right)$, so M is on the line. Now put $s = \frac{1}{2}\left(p + q + h + \frac{abc}{pq}\right)$. Then $s - m = \frac{1}{2}\left(q + \frac{abc}{pq}\right)$ and $\frac{abc}{p}(\bar{s} - \bar{m}) = \frac{abc}{2p}\left(\frac{1}{q} + \frac{pq}{abc}\right) = \frac{1}{2}\left(\frac{abc}{pq} + q\right)$, so S is on the Simson line of P, and by symmetry on that of Q. The two lines are not parallel, since $abc/p \neq abc/q$, so S is their intersection.
Two consequences come at once. The Simson lines of P and Q are perpendicular if and only if $abc/p = -abc/q$, that is, PQ is a diameter, and then $|s - \frac{h}{2}| = \frac{1}{2}$, so they meet on the nine-point circle.

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Example 2.
Let $ABCD$ be a convex cyclic quadrilateral in which neither AB nor CD is a diameter. The tangents at A and B meet at X, the tangents at C and D meet at Y, and the diagonals meet at P. Prove that X, P, Y are collinear.

Solution.
The standard way uses poles and polars (L5-GE1). If AB and CD meet at Q, then X and Y are the poles of AB and CD. As Q lies on both lines, La Hire puts X and Y on the polar of Q. By Brocard's theorem that polar is the line through P and $AD \cap BC$, so this line contains P. This needs the whole theory of polars, and the case $AB \parallel CD$ (and $AD \parallel BC$) has to be argued separately.
With complex numbers there is nothing to set up. By Theorem 2.1(2) and (3),

$$x = \frac{2ab}{a+b}, \qquad y = \frac{2cd}{c+d}, \qquad p = \frac{ac(b+d) - bd(a+c)}{ac - bd}.$$

Write $N = ab(c+d) - cd(a+b)$. Direct expansion gives

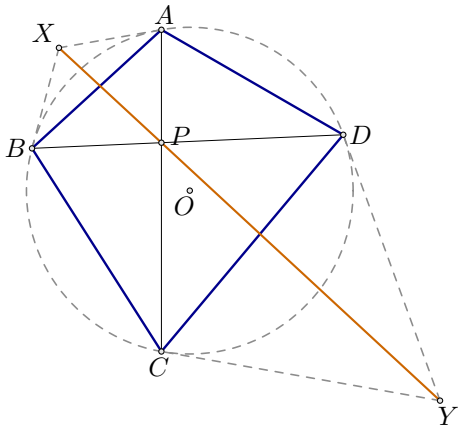
$$x - p = \frac{(a-b)N}{(a+b)(ac-bd)}, \qquad y - p = -\frac{(c-d)N}{(c+d)(ac-bd)}.$$

If $N = 0$ then $x = p$ and we are done. Otherwise

$$\frac{x-p}{y-p} = -\frac{a-b}{a+b} \cdot \frac{c+d}{c-d},$$

a product of two purely imaginary numbers, hence real. The denominators are nonzero, as $a+b \neq 0$ and $c+d \neq 0$ because the chords are not diameters, and $ac \neq bd$ because the diagonals meet. Parallel sides needed no special treatment.

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1.4 The incentre and the substitution $a = x^2, b = y^2, c = z^2$

The incentre is not a rational function of a, b, c. The bisector of angle A meets the circle at the midpoint of an arc BC, and the two arc midpoints are the two square roots of bc, up to sign. So we take square roots once and for all.

Theorem 4.1.
Let A, B, C lie on the unit circle. Then there are x, y, z on the unit circle with $a = x^2$, $b = y^2$, $c = z^2$ such that the midpoints of the arcs BC, CA, AB not containing A, B, C are $-yz$, $-zx$, $-xy$. With this choice the incentre is $i = -(xy + yz + zx)$, the excentre opposite A is $xy - yz + zx$, and the midpoint of arc BAC is yz.

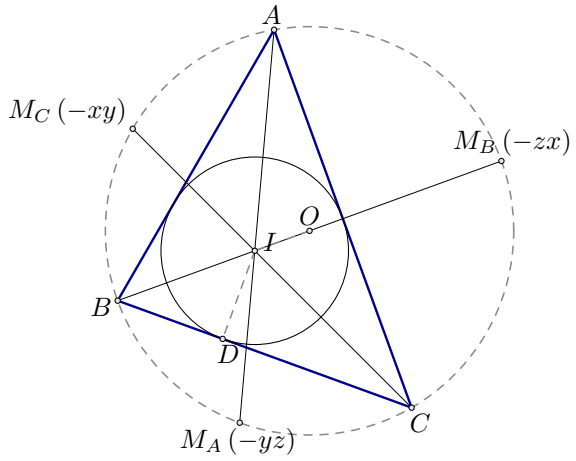
Proof. Label so that A, B, C are anticlockwise, with $a = e^{2i\alpha}$, $b = e^{2i\beta}$, $c = e^{2i\gamma}$ and $\alpha < \beta < \gamma < \alpha + \pi$. Put $x = e^{i\alpha}$, $y = -e^{i\beta}$, $z = e^{i\gamma}$. The arc AB not containing C runs from argument 2α to 2β, with midpoint $e^{i(\alpha+\beta)} = -xy$. In the same way the midpoint of arc BC is $e^{i(\beta+\gamma)} = -yz$, and arc CA runs from 2γ to $2\alpha + 2\pi$, with midpoint $e^{i(\alpha+\gamma+\pi)} = -zx$.

By the incentre-excentre lemma (L4-GE1), $M_B A = M_B I$ and $M_C A = M_C I$, so the line $M_B M_C$ is the perpendicular bisector of AI, and I is the reflection of A in that line. By Theorem 2.1(5),

$$i = m_b + m_c - m_b m_c \bar{a} = -zx - xy - \frac{x^2 yz}{x^2} = -(xy + yz + zx).$$

The same lemma says that M_A is the midpoint of II_A, so $i_A = 2m_A - i = xy - yz + zx$. The other arc midpoint of BC is $-m_A = yz$. ■

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Example 3.
Prove Euler's formula $OI^2 = R^2 - 2Rr$.

Solution.
Take $R = 1$ and use Theorem 4.1. The inradius r is the distance from I to BC, which is half the distance from I to its reflection $b + c - bc\bar{i}$. Here $\bar{i} = -\frac{x+y+z}{xyz}$, so $bc\bar{i} = -\frac{yz(x+y+z)}{x}$ and

$$i - (b + c - bc\bar{i}) = -\frac{x^2y + x^2z + xy^2 + xz^2 + y^2z + yz^2 + 2xyz}{x} = -\frac{(x+y)(y+z)(z+x)}{x}.$$

Hence $2r = |(x+y)(y+z)(z+x)|$. Let $w = \frac{(x+y)(y+z)(z+x)}{xyz}$. Its conjugate is $\frac{(x+y)(y+z)(z+x)}{x^2y^2z^2} \cdot xyz = w$, so w is real and $|w| = 2r$.
On the other hand $OI^2 = i\bar{i} = \frac{(xy+yz+zx)(x+y+z)}{xyz}$, and the identity $(x+y)(y+z)(z+x) = (x+y+z)(xy+yz+zx) - xyz$ gives $OI^2 = w + 1$. Since I is inside the circle, $OI^2 < 1$, so $w < 0$ and $w = -2r$. Therefore $OI^2 = 1 - 2r$, which is Euler's formula for $R = 1$. Scaling gives the general case. The sign argument at the end is the step a marker checks, since $|w| = 2r$ alone gives $OI^2 = 1 \pm 2r$.

1.5 When complex numbers are fast, and when they are hopeless

Complex numbers are fast when:

1. one circle passes through the main points, and every other point is a foot, a reflection, a tangent intersection, a chord intersection, a centroid or an orthocentre built from points on that circle,
2. the problem involves the incentre, excentres or arc midpoints, which Theorem 4.1 turns into polynomials,
3. there are regular polygons or rotations, which are multiplication by a root of unity,
4. the conclusion is a collinearity, concurrence, perpendicularity, a point on the circle or an equality of angles, each of which is the reality of one quotient.

They are slow or hopeless when there are two or more circles neither of which contains most of the points. When you need the circumcentre of three points not on the unit circle, or the intersection of two lines neither of which is a chord, since each such step roughly doubles the degree of

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the expressions. When the condition is about sums of lengths such as $AB + AC = 2BC$, which involve moduli and not polynomials, and for inequalities. A useful habit is to estimate degrees before starting. If the final identity will have degree above about ten in the variables, look for a synthetic lemma that replaces a hard point by an easy one. For example, the second intersection of line AI with the circle is $-yz$, which is far easier to use than the foot of the bisector on BC.

1.6 Writing a bash a marker will accept

1. State the set-up in one sentence, saying which circle is the unit circle and that each point is denoted by its lower-case letter. If you use Theorem 4.1, say which arc midpoints are $-xy$, $-yz$, $-zx$.
2. Quote the formulas of Theorem 2.1 by name. Derive any other formula you use in a line.
3. Write each point as an explicit expression before combining them, and show each conjugate you compute.
4. Reduce the claim to a single statement, such as a quotient being real or an expression vanishing, and show the algebra in factored form. "One can check that" earns nothing.
5. Justify every division, for example $a + b \neq 0$ because AB is not a diameter, $ab \neq cd$ because the lines are not parallel, and so on. If a factor you cancel can vanish, treat that case.
6. Finish. A computation abandoned halfway usually scores zero, because no intermediate line is a geometric statement a marker can credit.

Common traps.

- Using $\bar{z} = 1/z$ for a point not on the unit circle, such as H, I or a foot.
- Choosing x, y, z without the sign convention of Theorem 4.1, which swaps the incentre with an excentre.
- Dividing by $a + b$ when AB might be a diameter, or by $ab - cd$ when the chords might be parallel.
- Forgetting that the argument of a quotient gives a directed angle, so an equality of angles is only up to 180° unless the configuration is fixed.
- Starting a computation of degree twenty when one synthetic observation would cut the computation to degree five.

1.7 How to spot this in a contest

1. A triangle with its circumcircle and several feet of perpendiculars, reflections or tangents.
2. A cyclic quadrilateral together with intersections of its sides, diagonals or tangents.
3. Arc midpoints, the incentre and excentres together with the circumcircle.
4. A regular polygon, or equilateral triangles and squares erected on a figure.
5. A synthetic attempt that keeps needing a configuration case split. The algebra treats all cases at once.

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2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Let ABC be a triangle with circumcircle ω, and let D and E be distinct points of ω, different from A, B, C, such that $DE \parallel BC$. Prove that the Simson lines of D and E with respect to ABC meet on the altitude from A.

Problem 2 ⋆⋆ Let $A_1A_2\ldots A_n$ be a regular polygon with centre O, where $n \geq 3$, and let P be any point of its plane. Let F_k be the foot of the perpendicular from P to the line A_kA_{k+1}, where $A_{n+1} = A_1$. Prove that the centroid of $F_1, \ldots, F_n$ is the midpoint of OP.

Problem 3 ⋆⋆⋆ Let $ABCD$ be a convex cyclic quadrilateral with $\angle ACB = 30°$ and $\angle CAD = 45°$. The tangents to its circumcircle at A and B meet at X, the tangents at C and D meet at Y, and the diagonals meet at P. By Example 2, X, P, Y are collinear. Find XP/PY.

Problem 4 ⋆⋆⋆ Let A, B, C, D be four distinct points on a circle ω and let P be a point of ω different from all of them. Prove that the angle between the Simson line of P with respect to triangle ABC and the Simson line of P with respect to triangle ABD does not depend on P, and equals the angle between the lines AC and AD.

Problem 5 ⋆⋆⋆ Let ABC be a triangle with $\angle BAC \neq 90°$, incentre I and circumcircle ω. The tangents to ω at B and C meet at T. Let M_B and M_C be the midpoints of the arcs CA and AB of ω not containing B and C, and let the tangents at M_B and M_C meet at K. Prove that I, T, K are collinear.

Problem 6 ⋆⋆⋆⋆ Let P and Q be distinct points on the circumcircle of triangle ABC, different from A, B, C. Prove that the Simson lines of P and Q meet on the nine-point circle if and only if PQ is a diameter, or PQ is parallel to the Simson line of P, or PQ is parallel to the Simson line of Q.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO2 2011/12 Problem 1** ⋆⋆
   E is a chord intersection and the midpoints are averages. Compute the two circumcentres.
2. **BMO2 2006/07 Problem 3** ⋆⋆
   The 60 degree angle at A makes c equal to b times a cube root of unity. Then find P and Q on the chords AB and AC.
3. **IMO 2008 Problem 1** ⋆⋆
   With O = 0, compute the square of the distance from O to A1 by Pythagoras and check that this square is symmetric.
4. **BMO2 2015/16 Problem 3** ⋆⋆
   P and Q are chord intersections. Write each perpendicularity as a quotient being purely imaginary.

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5. **BMO2 2007/08 Problem 2** ⋆⋆⋆
   Use the x, y, z substitution. Each right or half-right angle at I is a condition on one quotient.

6. **USAMO 2010 Problem 1** ⋆⋆⋆
   Make AB a diameter of the unit circle. Each foot comes from Theorem 2.1(4).

7. **BMO2 2021/22 Problem 4** ⋆⋆⋆
   Put the circle as the unit circle with t a point of the circle. The tangent at T meets lines through B and C perpendicular to BC.

8. **IMO 2014 Problem 4** ⋆⋆⋆
   Each angle condition makes AP or AQ a chord from A to a point you can name. Then M and N are reflections of A.

9. **USAMO 2012 Problem 5** ⋆⋆⋆⋆
   Put P at 0 and the line through P on the real axis. Reflecting a line through 0 conjugates its direction.

10. **EGMO 2017 Problem 6** ⋆⋆⋆⋆
    Replace G and O by any point P and compute the second intersection of the circle through C and two reflections of P with the circumcircle.

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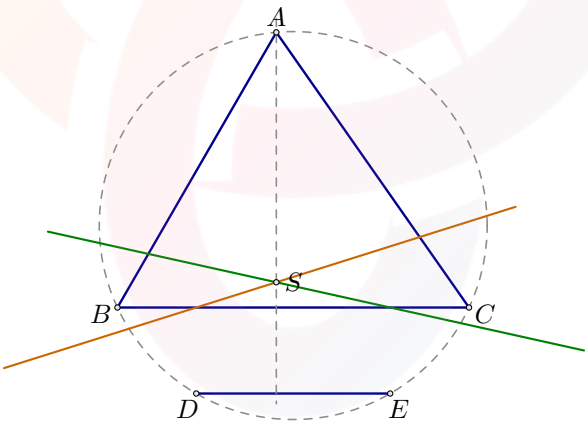
4 Hints

1. Use the intersection formula from Example 1 and Theorem 2.1(6).
2. Put the vertices at the nth roots of unity and add up the foot formula.
3. Take moduli in the formula for $\frac{x-p}{y-p}$ from Example 2, and compute $|a-b|/|a+b|$ from the central angle.
4. Example 1 gives the direction of a Simson line in the form $d/\bar{d}$.
5. Theorem 4.1 and Theorem 2.1(2). Factor $k-i$ and $t-i$. They share a factor.
6. By Example 1 the condition is $|p+q+k|=1$ for a certain k of modulus 1. When is the sum of three unit vectors a unit vector?

5 Solutions

Problem 1

Let ABC be a triangle with circumcircle ω, and let D and E be distinct points of ω, different from A, B, C, such that $DE \parallel BC$. Prove that the Simson lines of D and E with respect to ABC meet on the altitude from A.



Take ω as the unit circle. By Theorem 2.1(6), $DE \parallel BC$ means $de = bc$. By Example 1 the Simson lines of D and E meet at

$$s = \tfrac{1}{2}\left(d+e+a+b+c+\frac{abc}{de}\right) = a+\tfrac{1}{2}(b+c+d+e).$$

We show that S lies on the line through A perpendicular to BC. If $b+c+d+e=0$ then $s=a$ and we are done. Otherwise, using $de=bc$,

$$\overline{b+c+d+e} = \frac{b+c}{bc}+\frac{d+e}{de} = \frac{b+c+d+e}{bc}, \qquad \overline{b-c} = -\frac{b-c}{bc},$$

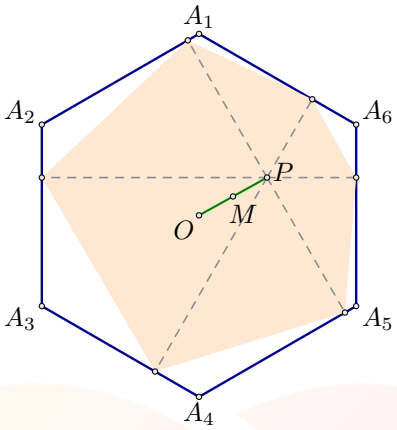
so the conjugate of $\frac{s-a}{b-c} = \frac{b+c+d+e}{2(b-c)}$ is $-\frac{b+c+d+e}{2(b-c)}$. The quotient is purely imaginary, so $AS \perp BC$ and S lies on the altitude from A.

Geometrically, $\frac{1}{2}(b+c)$ and $\frac{1}{2}(d+e)$ both lie on the common perpendicular bisector of the parallel chords, which passes through 0, so their sum does too. Translating by a moves that line onto the altitude.

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Problem 2

Let $A_1A_2\dots A_n$ be a regular polygon with centre O, where $n \geq 3$, and let P be any point of its plane. Let F_k be the foot of the perpendicular from P to the line A_kA_{k+1}, where $A_{n+1} = A_1$. Prove that the centroid of $F_1, \dots, F_n$ is the midpoint of OP.



Scale so that the circumcircle is the unit circle and $a_k = \zeta^k$ with $\zeta = e^{2\pi i/n}$. Theorem 2.1(4) holds for any point P, so

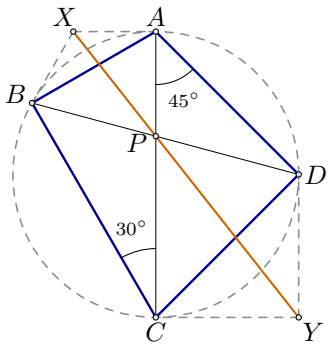
$$f_k = \tfrac{1}{2}\left(\zeta^k + \zeta^{k+1} + p - \zeta^{2k+1}\bar{p}\right).$$

Summing over $k = 1, \dots, n$ gives $\sum \zeta^k = 0$ and $\sum \zeta^{k+1} = 0$, because $\zeta \neq 1$ is an nth root of unity, and $\sum \zeta^{2k+1} = \zeta \sum (\zeta^2)^k = 0$, because ζ^2 is an nth root of unity different from 1 when $n \geq 3$. Hence $\sum f_k = \frac{n}{2}p$ and the centroid is $\frac{p}{2}$, the midpoint of OP.

For $n = 2$ the "polygon" is a segment traversed twice and $\zeta^2 = 1$, so the last sum is $-n$ and the statement fails, which is why $n \geq 3$ is needed.

Problem 3

Let $ABCD$ be a convex cyclic quadrilateral with $\angle ACB = 30^\circ$ and $\angle CAD = 45^\circ$. The tangents to its circumcircle at A and B meet at X, the tangents at C and D meet at Y, and the diagonals meet at P. By Example 2, X, P, Y are collinear. Find XP/PY.



Example 2 gives $\frac{XP}{PY} = \frac{|a-b|}{|a+b|} \cdot \frac{|c+d|}{|c-d|}$. If the central angle $\angle AOB$ is 2φ, then $|a-b| = 2\sin\varphi$ and $|a+b| = 2\cos\varphi$ (the diagonals of the rhombus with sides OA, OB), so $\frac{|a-b|}{|a+b|} = \tan\varphi$ for $\varphi < 90^\circ$.

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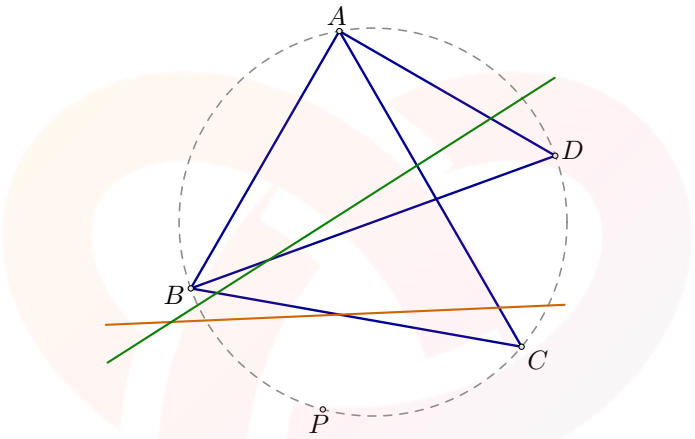
The vertices are in the order A, B, C, D round the circle, so C and D lie on the same arc AB, and $\angle ACB$ is the inscribed angle on the other arc AB. Hence $\angle AOB = 2\angle ACB = 60^\circ$ and $\varphi = 30^\circ$. In the same way $\angle COD = 2\angle CAD = 90^\circ$, and $\frac{|c+d|}{|c-d|} = \cot 45^\circ = 1$. Hence $XP/PY = \tan 30^\circ = \frac{\sqrt{3}}{3}$.

In general $XP : PY = \tan\angle ACB : \tan\angle CAD$, and a synthetic proof of this formula is a good exercise.

Answer: $\frac{\sqrt{3}}{3}$

Problem 4

Let A, B, C, D be four distinct points on a circle ω and let P be a point of ω different from all of them. Prove that the angle between the Simson line of P with respect to triangle ABC and the Simson line of P with respect to triangle ABD does not depend on P, and equals the angle between the lines AC and AD.



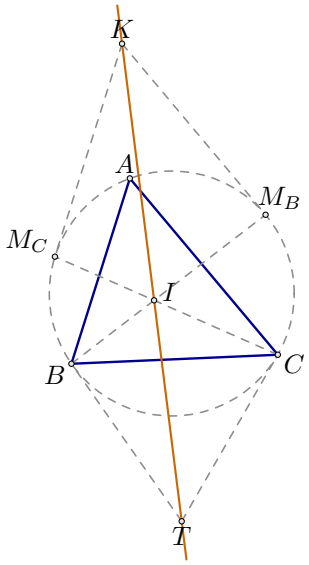
Take ω as the unit circle. A nonzero direction u is determined up to a real factor by $u/\bar{u}$, and if $u/\bar{u} = e^{2i\phi}$ then the line makes the angle ϕ (mod 180°) with the real axis. By Example 1 the Simson line of P for ABC has a direction u with $u/\bar{u} = abc/p$, and the Simson line for ABD has a direction v with $v/\bar{v} = abd/p$. If θ is the directed angle from the second line to the first, then $e^{2i\theta} = \frac{u/\bar{u}}{v/\bar{v}} = \frac{c}{d}$, which does not involve p.

For the lines AD and AC, the directions $c-a$ and $d-a$ satisfy $\frac{c-a}{\overline{c-a}} = -ac$ and $\frac{d-a}{\overline{d-a}} = -ad$ (from $\overline{c-a} = -\frac{c-a}{ac}$), so the directed angle θ' from AD to AC satisfies $e^{2i\theta'} = \frac{-ac}{-ad} = \frac{c}{d}$. Hence $\theta \equiv \theta'$ (mod 180°). The two Simson lines make the same angle as AD and AC, for every P. Since $C \neq D$, this angle is not 0, so in particular the Simson lines always meet.

Problem 5

Let ABC be a triangle with $\angle BAC \neq 90^\circ$, incentre I and circumcircle ω. The tangents to ω at B and C meet at T. Let M_B and M_C be the midpoints of the arcs CA and AB of ω not containing B and C, and let the tangents at M_B and M_C meet at K. Prove that I, T, K are collinear.

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Take ω as the unit circle and choose x, y, z as in Theorem 4.1, so $b = y^2$, $c = z^2$, $m_B = -zx$, $m_C = -xy$ and $i = -(xy + yz + zx)$. The points exist, since $b + c \neq 0$ as BC is not a diameter, and $m_B + m_C = -x(y + z) \neq 0$ since $y^2 \neq z^2$. By Theorem 2.1(2),

$$t = \frac{2y^2z^2}{y^2 + z^2}, \qquad k = \frac{2m_Bm_C}{m_B + m_C} = \frac{2x^2yz}{-x(y + z)} = -\frac{2xyz}{y + z}.$$

Let $N = x(y^2 + z^2) + yz(y + z)$. Expanding $(xy + yz + zx)(y + z) = x(y^2 + z^2) + 2xyz + yz(y + z)$ gives

$$k - i = \frac{(xy + yz + zx)(y + z) - 2xyz}{y + z} = \frac{N}{y + z}.$$

Similarly $(xy + yz + zx)(y^2 + z^2) + 2y^2z^2 = x(y + z)(y^2 + z^2) + yz(y + z)^2 = (y + z)N$, so

$$t - i = \frac{(y + z)N}{y^2 + z^2}.$$

Since T is outside ω and I is inside, $t \neq i$, so $N \neq 0$, and

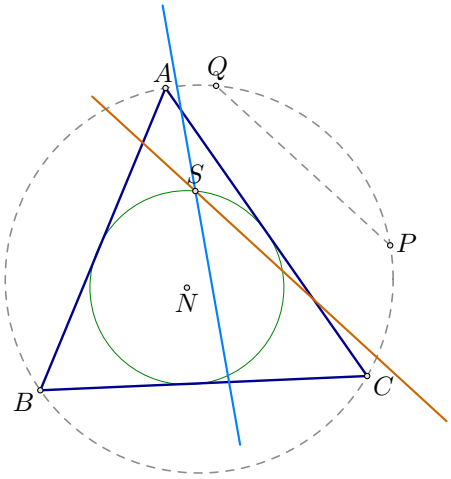
$$\frac{k - i}{t - i} = \frac{y^2 + z^2}{(y + z)^2}.$$

Its conjugate is $\frac{(y^2+z^2)/y^2z^2}{(y+z)^2/y^2z^2}$, the same number, so the quotient is real and I, T, K are collinear.

Problem 6

Let P and Q be distinct points on the circumcircle of triangle ABC, different from A, B, C. Prove that the Simson lines of P and Q meet on the nine-point circle if and only if PQ is a diameter, or PQ is parallel to the Simson line of P, or PQ is parallel to the Simson line of Q.

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Take the circumcircle as the unit circle. The Simson lines meet (Example 1, since $p \neq q$) at $s = \frac{1}{2}(p+q+h+k)$ with $k = \frac{abc}{pq}$, and $|k| = 1$. The nine-point circle has centre $\frac{h}{2}$ and radius $\frac{1}{2}$, so S lies on the circle if and only if $|p+q+k| = 1$.

Lemma. For u, v, w of modulus 1, $|u+v+w| = 1$ if and only if one of $u+v$, $v+w$, $w+u$ is 0.
Proof. If $u+v = 0$ then $|u+v+w| = |w| = 1$, and similarly in the other cases. Conversely, let $t = u+v+w$ with $|t| = 1$, and suppose $u+v \neq 0$. The points u and v lie on the unit circle and on the circle with centre $u+v$ and radius 1 (as $|u+v-u| = |v| = 1$). These two circles have different centres, so they meet in at most two points, and these are u and v. The points t and $-w$ lie on both circles as well, since $|t| = |-w| = 1$ and $(u+v) - t = -w$. Hence $-w \in \{u, v\}$, that is, $w+u = 0$ or $v+w = 0$. ■

Apply the lemma with $u = p$, $v = q$, $w = k$. The case $p+q = 0$ says PQ is a diameter. The case $p+k = 0$ says $p^2q = -abc$. By Example 1 the Simson line of P has direction d with $d/\bar{d} = abc/p$, and the chord PQ has direction $p-q$ with $(p-q)/\overline{(p-q)} = -pq$. Two directions are parallel exactly when these quotients are equal, that is, when $abc/p = -pq$, which is $p^2q = -abc$. So the case $p+k = 0$ says that PQ is parallel to the Simson line of P, and the case $q+k = 0$ says the same for Q. Since every step was an equivalence, this proves both directions.